IT5008: Tutorial 8 — Functional Dependencies

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AY26/27 S1



Example

Example Relation, $R = \{A, B, C\}$ and $\Sigma = \{\{A\} \rightarrow \{B\}, \{B\} \rightarrow \{C\}\}$

The column in red decides the color of the row: The rows having same values in the red column, have same color.

Task: Check which other columns are following "same value if same color" for each deciding (red) column. The columns for which this holds, are implied by the red column.

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$\{A\} \rightarrow \{A, B, C\}$

A	**B**	C
a1	b1	c1
a1	b1	c1
a2	b2	c2
a2	b2	c2
a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
a5	b2	c2

$\{B\} \rightarrow \{B, C\}$

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a2	b2	c2
a3	b3	c2
a3	b3	c2
a4	b1	c1
a4	b1	c1
a5	b2	c2

$\{B\} \rightarrow \{B, C\}$

A	B	**C**
a1	b1	c1
a1	b1	c1
a2	b2	c2
a2	b2	c2
a3	b3	c2
a3	b3	c2
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A	**B**	C
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a3	b3	c2
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$\{B\} \rightarrow \{B, C\}$

A	B	**C**
a1	b1	c1
a1	b1	c1
a2	b2	c2
a2	b2	c2
a3	b3	c2
a3	b3	c2
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a4	b1	c1
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$\{C\} \rightarrow \{C\}$

Scenario

Your company, *Apasaja Private Limited*, is commissioned by an online company offering several services to design the relational schema for the management of their users' profiles. A service is fully described and identified by its name. Each user can register to one or more services. A user is uniquely identified by her email as well as by her mobile number. Each user has both a postal address and a country of residence. The postal address, however, unambiguously identifies the country in which it is located. There can be several users with the same address.

We are given the following abstract schema for this application.

- $R = \{A, B, C, D, E\}$
- $\Sigma = \{\, \{A\} \to \{A, B, C\},\ \{A, B\} \to \{A\},\ \{B, C\} \to \{A, D\},\ \{B\} \to \{A, B\},\ \{C\} \to \{D\} \,\}$

Questions

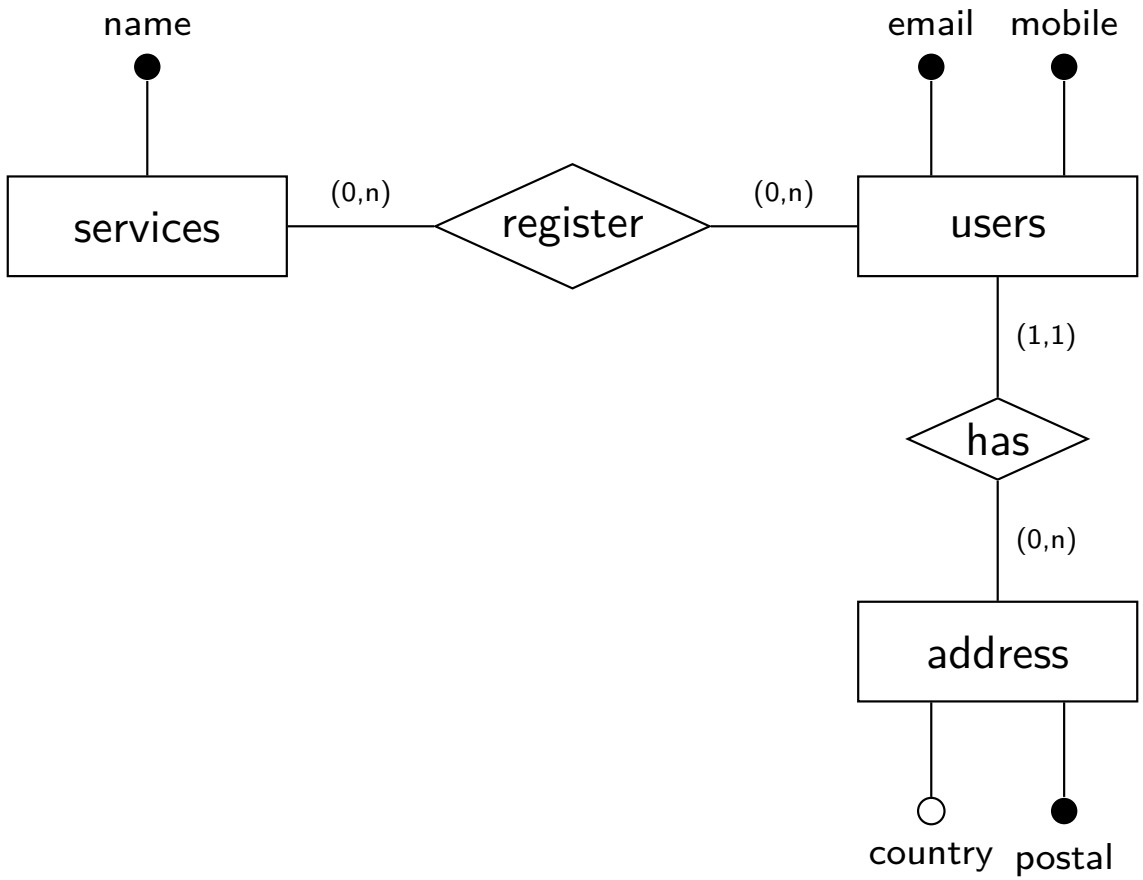
Preliminary *(not discussed in class)*

1. **Schema.**
   (a) Draw the entity-relationship diagram.
   (b) Translate the entity-relationship diagram into a schema.
   (c) From Σ, the text, the ER diagram and the schema, figure out the mapping of the attributes and the letters.

Discussion

2. **Calculations.**
   (a) Compute the attribute closures of the subsets of attributes of R with Σ in order to find the candidate keys of R with Σ.
   (b) Find the prime attributes of R with Σ.
3. **Basis or Cover.**
   (a) Compute one possible minimal cover (a.k.a. minimal basis) of R with Σ.
   (b) Compute one possible canonical cover of R with Σ.

1(a). Entity-Relationship Diagram



Filled circles are keys: `email` and `mobile` each identify a user; `postal` identifies an address (and so its `country`).

1(b). Translating to a Schema

One possible answer:

```
CREATE TABLE services (
  name VARCHAR(256) PRIMARY KEY
);

CREATE TABLE address (
  postal VARCHAR(10) PRIMARY KEY,
  country VARCHAR(100) NOT NULL
);
```

```
CREATE TABLE users (
  email VARCHAR(100) PRIMARY KEY,
  mobile NUMERIC UNIQUE NOT NULL,
  postal VARCHAR(10)
    REFERENCES address(postal)
);

CREATE TABLE register (
  name VARCHAR(256)
    REFERENCES services(name),
  email VARCHAR(100)
    REFERENCES users(email),
  PRIMARY KEY (name, email)
);
```

- has is $(1, 1)$ on the users side, so it folds into users as a foreign key.
- mobile is a second key: UNIQUE NOT NULL.

1(c). Mapping Attributes to Letters

Your company, Apasaja Private Limited, is commissioned by an online company offering several services to design the relational schema for the management of their users' profiles. A service is fully described and identified by its name. Each user can register to one or more services. A user is uniquely identified by her **email** as well as by her **mobile number**. Each user has both a postal address and a country of residence. The **postal address**, however, unambiguously identifies the **country** in which it is located. There can be several users with the same address.

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The service name neither depends on anything nor implies anything. Thus E is the service name.

1(c). The Mapping

Attribute	**Letter**
`email` / `mobile`	A / B (or vice versa)
`postal`	C
`country`	D
`name` (of service)	E

$R = \{A, B, C, D, E\}$ is one big relation: (*email*, *mobile*, *postal*, *country*, *service*) — one row per user *per service* they registered to.

2(a). Finding the Candidate Keys

Observation: E does not appear on the right-hand side of *any* functional dependency in Σ. So nothing can derive E: it must be in **every** candidate key.

We only need closures of supersets of $\{E\}$:

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- **Cardinality 2**
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- **Cardinality 3**, superset of $\{E\}$, but not a superset of a known candidate key
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Candidate keys: $\{A, E\}$ and $\{B, E\}$.

2(a). Sanity Check: All 31 Closures

Computing every closure by hand is tedious — the shortcut above saves most of the work. For reference, superkeys in red; candidate keys underlined.

$\{A\}^+ = \{A, B, C, D\}$
$\{B\}^+ = \{A, B, C, D\}$
$\{C\}^+ = \{C, D\}$
$\{D\}^+ = \{D\}$
$\{E\}^+ = \{E\}$
$\{A, B\}^+ = \{A, B, C, D\}$
$\{A, C\}^+ = \{A, B, C, D\}$
$\{A, D\}^+ = \{A, B, C, D\}$
$\underline{\{A, E\}^+ = \{A, B, C, D, E\}}$
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$\{A, B, C, D, E\}^+ = \{A, B, C, D, E\}$

Every red set contains E — as predicted.

Here's some help: compute closures yourself at functionaldepcheckeronline.streamlit.app

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Here the prime attributes are $\{A, B, E\}$ — email, mobile and service name.

3(a). Minimal Cover — Steps 0 & 1

Initial: start with Σ.

$$\{A\} \rightarrow \{A, B, C\}$$
$$\{A, B\} \rightarrow \{A\}$$
$$\{B, C\} \rightarrow \{A, D\}$$
$$\{B\} \rightarrow \{A, B\}$$
$$\{C\} \rightarrow \{D\}$$

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$$\{A, B\} \to \{A\}$$
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$$\{B\} \to \{A, B\}$$
$$\{C\} \to \{D\}$$

Step 1: Simplify the right-hand sides — one attribute on each right.

$$\{A\} \to \{A\} \quad \text{(split)}$$
$$\{A\} \to \{B\} \quad \text{(split)}$$
$$\{A\} \to \{C\} \quad \text{(split)}$$
$$\{A, B\} \to \{A\}$$
$$\{B, C\} \to \{A\} \quad \text{(split)}$$
$$\{B, C\} \to \{D\} \quad \text{(split)}$$
$$\{B\} \to \{A\} \quad \text{(split)}$$
$$\{B\} \to \{B\} \quad \text{(split)}$$
$$\{C\} \to \{D\}$$

3(a). Minimal Cover — Step 2

Step 2: Simplify the left-hand sides — drop an attribute if the rest still determines the right-hand side.

$\{A\} \to \{A\}$

$\{A\} \to \{B\}$

$\{A\} \to \{C\}$

$\{A, \cancel{B}\} \to \{A\}$ since $\{A\}^+ = \{A, B, C, D\} \ni A$

$\{B, \cancel{C}\} \to \{A\}$ since $\{B\}^+ = \{A, B, C, D\} \ni A$

$\{B, \cancel{C}\} \to \{D\}$ since $\{B\}^+ = \{A, B, C, D\} \ni D$

$\{B\} \to \{A\}$

$\{B\} \to \{B\}$

$\{C\} \to \{D\}$

We could also have reduced $\{B, C\} \to \{D\}$ to $\{\cancel{B}, C\} \to \{D\}$, since $\{C\} \to \{D\}$. Different choices can lead to different (equally valid) minimal covers.

3(a). Minimal Cover — Step 3

Step 3: Remove redundant FDs.

$\cancel{\{A\} \rightarrow \{A\}}$ (trivial)
$\{A\} \rightarrow \{B\}$
$\{A\} \rightarrow \{C\}$
$\cancel{\{A\} \rightarrow \{A\}}$ (trivial)
$\{B\} \rightarrow \{A\}$
$\cancel{\{B\} \rightarrow \{D\}}$ (redundant)
$\cancel{\{B\} \rightarrow \{A\}}$ (duplicate)
$\cancel{\{B\} \rightarrow \{B\}}$ (trivial)
$\{C\} \rightarrow \{D\}$

$\{B\} \rightarrow \{D\}$ is redundant:
$B \rightarrow A \rightarrow C \rightarrow D$.

3(a). Minimal Cover — Step 3

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$\{C\} \to \{D\}$

$\{B\} \to \{D\}$ is redundant:
$B \to A \to C \to D$.

One minimal cover

$$\left\{ \begin{matrix} \{A\} \to \{B\},\ \{A\} \to \{C\}, \\ \{B\} \to \{A\},\ \{C\} \to \{D\} \end{matrix} \right\}$$

3(a). Minimal Cover — Step 3

Step 3: Remove redundant FDs.

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$\{C\} \to \{D\}$

$\{B\} \to \{D\}$ is redundant:
$B \to A \to C \to D$.

One minimal cover

$$\begin{Bmatrix} \{A\} \to \{B\}, & \{A\} \to \{C\}, \\ \{B\} \to \{A\}, & \{C\} \to \{D\} \end{Bmatrix}$$

Another minimal cover

$$\begin{Bmatrix} \{A\} \to \{B\}, & \{B\} \to \{A\}, \\ \{B\} \to \{C\}, & \{C\} \to \{D\} \end{Bmatrix}$$

Here C is reached from B instead of A — fine, since A and B determine each other.

3(b). Canonical Cover

Canonical (compact minimal) cover

Just merge the dependencies of a minimal cover that share a left-hand side:

$$\{\, \{A\} \rightarrow \{B, C\},\ \{B\} \rightarrow \{A\},\ \{C\} \rightarrow \{D\} \,\}$$

One Algorithm to Rule Them All

The attribute closure algorithm is the most important algorithm for functional dependencies and normalization. Call it $C^*(X, \Sigma)$: the closure of the attributes X under the FDs Σ.

- **Q2(a) — keys:** for each $X \subseteq R$, compute $\varphi = C^*(X, \Sigma)$.
  X is a superkey iff $\varphi = R$; a candidate key if, in addition, no proper subset is.
- **Q3(a) Step 2 — shrink a left side:** to drop $B \in X$ from $X \to \{A\}$, compute $\varphi = C^*(X - \{B\},\ \Sigma)$; drop B iff $A \in \varphi$.
- **Q3(a) Step 3 — drop an FD:** to drop $X \to \{A\}$ from Σ, compute $\varphi = C^*(X,\ \Sigma - \{X \to \{A\}\})$; drop it iff $A \in \varphi$.

Tip: for any FD or normalization question, always start by computing (i) the **candidate keys** and (ii) a **minimal cover**. They simplify everything after.

Questions?
Drop a mail at: pratik.karmakar@u.nus.edu